AQ8.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Which of these sets form an orthonormal basis of R^3 R3 with respect to the dot product as the inner product in R^3 R3?

- [ ] $\{(1,0,0),(0,1,0)\}$ $\{(1,0,0),(0,1,0)\}$
- [ ] $\{(2,0,0),(0,2,0),(0,0,2)\}$ $\{(2,0,0),(0,2,0),(0,0,2)\}$
- [ ] $\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$ $\{(\sqrt{6}, 1, \sqrt{6}, 2), (\sqrt{11}, -3, \sqrt{11}, 1), (\sqrt{66}, 1, \sqrt{66}, 7, \sqrt{66}, -4)\}$
- [ ] $\{(2,1,-1),(-1,-1,1),(1,2,-2)\}$ $\{(2,1,-1),(-1,-1,1),(1,2,-2)\}$
- [ ] $\{(\frac{1}{\sqrt{2}},0,\frac{-1}{\sqrt{2}}),(0,1,0),(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}})\}$ $\{(\sqrt{2}, 1, 0, \sqrt{2}), (0, 1, 0), (\sqrt{2}, -1, 0, \sqrt{2}, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$ $\{(\sqrt{6}, 1, \sqrt{6}, 2), (\sqrt{11}, -3, \sqrt{11}, 1), (\sqrt{66}, 1, \sqrt{66}, 7, \sqrt{66}, -4)\}$

$\sqrt{}$
1, 6
$\sqrt{}$
1, 6
$\sqrt{}$
2), (11
$\sqrt{}$
−3, 11
$\sqrt{}$
1, 11
$\sqrt{}$
1), (66
$\sqrt{}$
1, 66
$\sqrt{}$
7, 66
$\sqrt{}$
−4)}
$\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$ {(2
$\sqrt{}$
1, 0, 2
$\sqrt{}$
−1), (0, 1, 0), (2
$\sqrt{}$
1, 0, 2
$\sqrt{}$
1)}

1 point

Consider two orthogonal vectors a, b in R^2. If $a + b$ and $a - b$ are orthogonal, then choose the correct option.

- $\|a\| = \|b\| = 1$
- $\|a\| = \|b\|$
- $\|a\| = 2\|b\|$
- $2\|a\| = \|b\|$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\|a\| = \|b\|$

1 point

Consider $a = (1, 1), b = (1, -1)$. Let $V = Span\{a, b\}$. Choose the correct options by considering the standard inner product (dot product).

- [ ] Vectors a, b form an orthogonal basis for V.
- [ ] Vectors a, b form an orthonormal basis for V.
- [ ]

There exist scalar multiples of a, ba, b which form an orthonormal basis

There do not exist scalar multiples of a, ba, b which form an orthonormal basis

No, the answer is incorrect.

Score: 0

Accepted Answers:

Vectors a, ba, b form an orthogonal basis for VV .

There exist scalar multiples of a, ba, b which form an orthonormal basis

Level 2

1 point

Choose the set of correct statements.

The determinant of a matrix formed by 33 orthonormal vectors in R^3R3 is ±1±1.

The determinant of a matrix formed by 33 orthonormal vectors in R^3R3 is 00.

The determinant of a matrix formed by 22 orthonormal vectors in R^2R2 is ±1±1.

The determinant of a matrix formed by 22 orthonormal vectors in R^2R2 is 00.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of a matrix formed by 33 orthonormal vectors in R^3R3 is ±1±1.

The determinant of a matrix formed by 22 orthonormal vectors in R^2R2 is ±1±1.

1 point

Consider a system of linear equations:

$$2x_1 + 2x_2 + 7x_3 = b_1 \text{ 2x1 + 2x2 + 7x3 = b1}$$

$$2x_1 + x_2 - 10x_3 = b_2 \text{ 2x1 + x2 − 10x3 = b2}$$

$$3x_1 - 2x_2 + 2x_3 = b_3 \text{ .3x1 − 2x2 + 2x3 = b3.}$$

Let AA be the coefficient matrix of the given system of linear equations. Let a matrix BB contain the column vectors of AA, which are normalized by their respective norms, as its columns (i.e. first column vector of AA normalized by its norm is the first column of BB). Which of the following statements are true ?

The determinant of BB^T BBT is 11.

BB^T BBT is an identity matrix.

BB^T BBT is a scalar matrix.

BB^T BBT is a diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of BB^T BBT is 11.

BB^T BBT is an identity matrix.

BB^T BBT is a scalar matrix.
BB^T BBT is a diagonal matrix.
1 point
Choose the correct option(s).
- [ ] The vectors in an orthonormal set are linearly independent.
- [ ] A set of linearly dependent vectors can be orthonormal.
- [ ] A set of linearly independent vectors is always orthonormal.
- [ ] A set of linearly independent vectors is always orthogonal but not orthonormal.
- [ ] The vectors in an orthogonal set are linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The vectors in an orthonormal set are linearly independent.
The vectors in an orthogonal set are linearly independent.

Consider the inner product $\langle u, v\rangle = 3u_1 v_1 + 2u_2 v_2$ $\langle$u, v$\rangle$ = 3u1v1 + 2u2v2 where $u = (u_1, u_2), v = (v_1, v_2)$u = (u1, u2), v = (v1, v2) are vectors in R^2 R2. Answer questions 7 and 8 based on this information.
1 point
Which of the following is an orthonormal basis with respect to the inner product defined above?
○
$\{(2,3),(-4,4)\}$ $\{(2,3),(-4,4)\}$
○
$\{\frac{1}{\sqrt{30}}(2,3),\frac{1}{\sqrt{80}}(-4,4)\}$ $\{\frac{1}{\sqrt{30}}(2,3),\frac{1}{\sqrt{80}}(-4,4)\}$
○
$\{\frac{1}{\sqrt{13}}(2,3),\frac{1}{\sqrt{32}}(-4,4)\}$ $\{\frac{1}{\sqrt{13}}(2,3),\frac{1}{\sqrt{32}}(-4,4)\}$
○
$\{(2,3),(-3,2)\}$ $\{(2,3),(-3,2)\}$
○
$\{\frac{1}{\sqrt{13}}(2,3),\frac{1}{\sqrt{13}}(-3,2)\}$ $\{\frac{1}{\sqrt{13}}(2,3),\frac{1}{\sqrt{13}}(-3,2)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{\frac{1}{\sqrt{30}}(2,3),\frac{1}{\sqrt{80}}(-4,4)\}$ $\{\frac{1}{\sqrt{30}}(2,3),\frac{1}{\sqrt{80}}(-4,4)\}$

1(−4, 4)}

1 point

Use the orthonormal basis $\{u, v\}$ {u, v} obtained in question 7 with respect to the defined inner product. Express the vector (4, 0)(4, 0) as a linear combination of the basis vectors uu and vv, as (4, 0) = $c_1 u$ + $c_2 v$(4, 0) = c1u + c2v. Which of the following gives the coefficients of the linear combination?

○

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

−48

○

$c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}}$c1 = 13

√

24, c2 = 32

√

−48

○

$c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}}$c1 = 13

√

8, c2 = 32

√

−16

○

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

48

No, the answer is incorrect.

Score: 0

Accepted Answers:

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

−48

Check Answers

Your score is: 0/8